

UMESH KRISHNA KAMISSETTY

Data Engineer

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PROFESSIONAL SUMMARY

- 5+ years of experience designing, implementing, and managing enterprise-scale data engineering platforms across AWS, Azure, and Databricks, handling multi-terabyte transactional, analytical and streaming workloads.
- Engineered batch and streaming ETL pipelines using PySpark, Glue, ADF, Kafka, Kinesis, Databricks Structured Streaming, and SQL-based transformations, achieving sub-minute latency and reliable data ingestion for billions of daily records.
- Architected and optimized data warehouses, lake houses, and hybrid cloud/on-prem infrastructures leveraging Snowflake, Redshift, Synapse, Delta Lake, SQL Server, and ADLS, improving query performance by up to 50% and reducing compute/storage costs by 35%.
- Implemented CI/CD pipelines, Infrastructure-as-Code (IaC), and containerized workflows using Jenkins, Git, Docker, Terraform, Code Pipeline, AKS, EKS, and Lambda, enabling automated deployments, reproducibility, and multi-environment reliability exceeding 99%.
- Troubleshoot and resolved complex production issues across distributed data environments, including Spark job optimization, multi-AZ streaming pipelines, cloud networking, ML feature pipelines, and RAG/NLP pipelines, enhancing pipeline SLA adherence by 95% and minimizing operational downtime.

TECHNICAL SKILLS

Programming Scripting	& Python, PySpark, SQL, Shell scripting
Cloud Platforms – AWS	S3, Glue, EMR, Redshift, Lake Formation, Step Functions, Lambda, RDS, CloudWatch, EC2, VPC, SageMaker, EKS, ECS, CodePipeline, OpenSearch, Migration Hub
Cloud Platforms – Azure	Azure Data Factory, ADLS Gen2, Azure Synapse Analytics, Azure SQL, Azure ML, Azure Functions, AKS, Azure Blob Storage, Azure VM, Azure VNet, Migration Hub
Data Warehousing & Analytics	Snowflake, SQL Server, Synapse, Redshift, Databricks Delta Lake, SSMS, SSIS, SSAS, SSRS
ETL & Data Integration	ETL (Batch & Streaming), DBT, Airflow, ADF, Glue Jobs, PySpark Transformations, Kafka, Kinesis
Big Data & Distributed Processing	Hadoop, Hive, Spark, PySpark, MapReduce, EMR, Delta Lake
Machine Learning & AI	SageMaker, Azure ML, AI Platform, NLP pipelines, RAG pipelines, LLM/Chatbot integration, Agentic LLM-assisted coding, Predictive modeling, Feature engineering pipelines

Data Modeling & Governance	Schema design, Data modeling, Star/Snowflake schemas, CDC pipelines, Data lineage, Access control, Data quality monitoring
Visualization & BI	Tableau, Power BI, Looker
CI/CD & DevOps	Git, Jenkins, CodePipeline, Azure DevOps, Docker, Terraform (IaC), Containerization, Automated testing, Pipeline orchestration
Monitoring & Observability	CloudWatch, Azure Monitor, Log Analytics, Databricks Jobs API, Custom Python telemetry, SLA tracking
Database Migration & Optimization	SQL Server → Cloud migrations, Oracle → Cloud migrations, Query optimization, Partitioning & indexing, Workload isolation
Networking & Security	VPC design, Network isolation, IAM, Role-based access, KMS encryption, End-to-end secure pipelines
Real-Time / Streaming Data	Kafka, Kinesis, Spark Structured Streaming, Event Hubs, Low-latency processing pipelines
Other Technologies / Frameworks	Docker, EKS, AKS, ECS, Lambda, Azure Functions, Cloud Functions, SageMaker endpoints

PROFESSIONAL EXPERIENCE

First United Bank

Aug 2025 - Present

Data Engineer II

Seattle, WA

- Architected and delivered a cloud-native financial data lakehouse on AWS (S3, Glue, EMR, Redshift, Databricks Delta), ingesting 3.2+ billion transactions daily and enabling low-latency risk, compliance, and regulatory analytics through fully ACID-compliant pipelines supporting enterprise-scale workloads.
- Led migration of 80+ TB of mission-critical SQL Server and Oracle databases to AWS S3 and Databricks Delta, implementing robust CDC, schema evolution, and automated reconciliation pipelines to achieve zero-downtime cutover and 100% data validation accuracy.
- Designed and implemented enterprise-grade streaming architecture using Kafka, Kinesis, and Spark Structured Streaming with exactly-once guarantees and event-time processing, reducing ingestion latency by 72% while eliminating data skew across high-volume financial feeds.
- Implemented secure, fine-grained data governance using AWS Lake Formation, IAM, VPC endpoint isolation, and KMS encryption, enforcing column-level access controls across sensitive financial datasets and ensuring full compliance with SEC and FINRA regulatory standards.
- Built ML feature engineering pipelines on Databricks powering 18+ production risk, fraud, and portfolio optimization models, integrating SageMaker endpoints with sub-35ms online feature serving and scalable feature caching.

- Provisioned and standardized cloud infrastructure using Terraform, orchestrating 18+ Databricks clusters, 180+ EMR workloads, automated S3 lifecycle policies, Lambda services, and KMS encryption, ensuring reproducibility and compliance across multi-environment deployments.
- Optimized large-scale Databricks Delta Lake tables using Z-order clustering, auto-compaction, and file pruning, improving analytical query throughput by 70% across multi-terabyte datasets while reducing cluster compute costs by 35%.
- Tuned complex PySpark workloads leveraging Adaptive Query Execution (AQE), dynamic partition pruning, and optimized join strategies, reducing compute consumption by 48% while consistently meeting SLA-driven processing timelines.
- Developed near-real-time ingestion pipelines using Kinesis Data Streams, Lambda, and Spark Structured Streaming, processing 10+ million trade and market events per minute, enabling sub-3-minute dashboard refresh cycles for portfolio and risk stakeholders.
- Designed modular DBT transformation layers in Databricks SQL and PySpark, governing 800+ curated datasets, capturing full data lineage, and implementing automated regression testing to improve analytical reliability and deployment confidence.
- Integrated AWS OpenSearch and Kibana dashboards to enable real-time anomaly detection across trade, risk, and operational datasets, reducing mean incident response time by 55%.
- Automated CI/CD pipelines using Jenkins, Git, and Docker, embedding unit and integration testing, schema validation, and orchestrated Databricks deployments to achieve 99.9% deployment reliability and accelerated release velocity.
- Mentored and led a team of 6 engineers, institutionalizing Spark optimization standards, Delta Lake governance practices, and CI/CD frameworks—accelerating enterprise data delivery by 35%.

United Health Group

March 2024 - Aug 2025

Data Engineer I

Dallas, TX

- Architected hybrid cloud and on-prem analytics platform integrating Azure Data Factory, ADLS Gen2, Azure Synapse, and on-prem SQL Server, processing 350+ million clinical and supply chain records daily with enterprise-grade reliability and governance controls.
- Built enterprise Retrieval-Augmented Generation (RAG) pipelines indexing 20+ million regulatory and compliance documents using vector embeddings and similarity search, enabling high-accuracy knowledge retrieval for audit and compliance operations.
- Developed LLM-powered internal knowledge assistants integrated with Azure Cognitive Search and on-prem repositories, reducing enterprise data discovery time by 75% and significantly improving decision turnaround for clinical and regulatory teams.
- Implemented real-time streaming pipelines using Azure Event Hubs and Kafka, achieving sub-60-second end-to-end latency for manufacturing, inventory, and operational analytics.
- Designed modular NLP pipelines using Python and PySpark to extract entities and normalize unstructured clinical and regulatory text, scaling processing to 80+ GB per day while improving extraction accuracy and standardization.
- Developed enterprise-grade ELT pipelines using Airflow, Azure Data Factory, SSIS, and DBT, standardizing transformation frameworks and enabling automated regression testing across 600+ curated datasets.
- Containerized ML and ETL workloads for AKS and Azure Functions, implementing autoscaling, secure service-to-service authentication, and hybrid networking to support resilient multi-environment deployments.
- Built comprehensive monitoring and observability frameworks using Azure Monitor, Log Analytics, and custom telemetry, proactively detecting 98% of pipeline anomalies before downstream business impact.

- Automated agentic LLM-assisted code generation for ETL and transformation pipelines, producing reusable PySpark and SQL templates and accelerating feature delivery by 40%.
- Implemented CI/CD pipelines with Azure DevOps, Jenkins, and Git, enforcing automated testing, schema validation, and controlled multi-environment deployments, achieving 99.7% release stability.
- Mentored cross-functional engineering teams in distributed data architecture, Spark optimization, and hybrid cloud integration, improving overall project delivery velocity by 35%.

Nationwide Insurance

August 2020 - July 2023

Data Engineer I

Bangalore, India

- Architected a multi-tier AWS cloud data Lakehouse (S3, Glue, EMR, Snowflake) ingesting 750+ million insurance policy, claims, and customer records daily, enabling scalable enterprise analytics across underwriting, actuarial, and risk domains.
- Optimized enterprise warehouse performance across Redshift and Snowflake using distribution strategies, sort keys, clustering, and advanced indexing, reducing query latency by 45% while supporting 200+ concurrent analytical users.
- Designed Snowflake table architectures leveraging micro-partitioning, clustering keys, and materialized views, improving performance by 65% on multi-terabyte actuarial and claims datasets.
- Reduced overall ETL and Snowflake warehouse costs by 40% through workload isolation, query caching strategies, warehouse right-sizing, and prioritized execution for high-value analytical workloads.
- Containerized Spark and supporting services using Docker and deployed on EKS for elastic scaling and reproducible workloads, increasing pipeline throughput by 50% during peak processing windows.
- Developed scalable batch ETL pipelines with AWS Glue (PySpark), implementing incremental processing, schema evolution, and automated data quality validations, reducing pipeline runtimes by 55%.
- Orchestrated batch and streaming workflows using AWS Step Functions and Lambda with SLA-aware retry logic and event-driven triggers, improving job reliability and execution success rates to 99.8%.
- Built centralized observability frameworks using CloudWatch and Snowflake resource monitors, proactively detecting pipeline failures and resource overutilization, reducing incident resolution time by 45%.
- Implemented CI/CD pipelines with Code Pipeline, Jenkins, and Git, automating deployment of Glue jobs, Snowflake DDL/DML scripts, and Python libraries, achieving 99.5% deployment stability across environments.
- Partnered with actuarial, claims, and analytics stakeholders to deliver 80+ Tableau and Power BI dashboards, reducing manual reporting effort by 70% and accelerating executive decision-making cycles.

EDUCATION

- Master's in information systems | Cleveland State University
- Bachelor's in computer science | JNTU Ananthapur